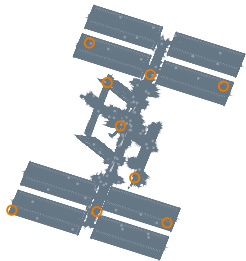


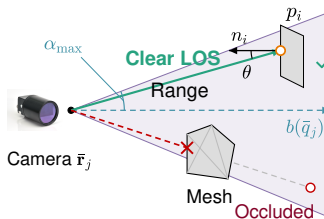
Viability-preserving rollout ADP for orbital inspection

a Mission inputs



○ Required targets $\mathcal{K}$

b Camera and LOS



Range · FOV · incidence · clear LOS

Valid targets $\rightarrow G_j$

$$s = (j, m, b, h)$$

c HCW transfer $\rightarrow$ SOOA graph

1 Solve HCW

Rest-to-rest · 90 s
C70 $\rightarrow$ C68
Trajectory + control
 $\mathbf{x}_{ij}(t), \mathbf{u}_{ij}(t)$

2 Audit motion

- ✓ Input + speed
- ✓ Mesh clearance
- ✓ Terminal errors

All pass: $\chi_{ij} = 1$

Fail $\rightarrow$ omit edge

3 Build graph

SOOA a_{ij}

C70 $\xrightarrow{\ell_{ij}}$ C68
Source i View j

Edge: motion + cost

Node: view + mask G_j

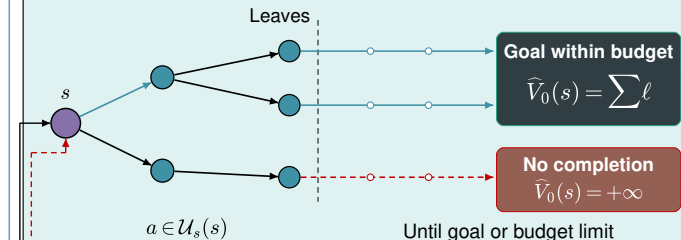
d Rollout ADP

1 Exact lookahead

Safe actions · depth d

2 Full policy tails

Deterministic greedy μ



3 Bellman backup · discard $+\infty$ actions

$$\widehat{Q}_d(s, a) = \ell_{ja} + \widehat{V}_{d-1}(f(s, a))$$

State update and replan

e Replanning

First action a^*

Minimum finite $\widehat{Q}_d(s, a)$

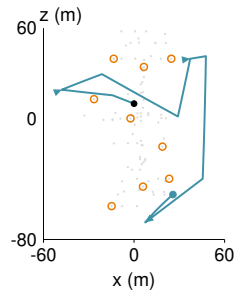
Append arc + update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

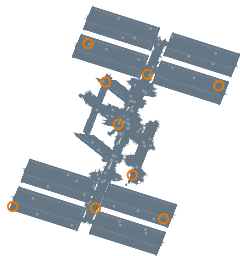
Planned route



Required targets: 9/9

Orbital inspection: sensing, audited transfers and rollout ADP

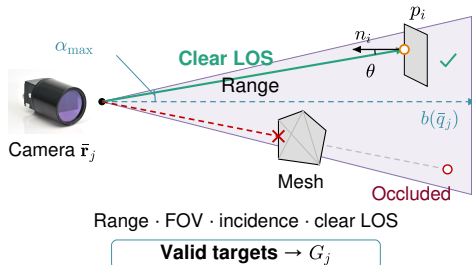
a Mission



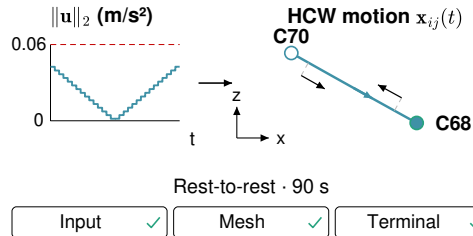
○ Required targets $\mathcal{K}$

Initial state; budget H

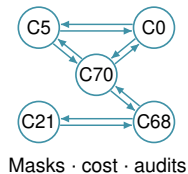
b Camera and LOS



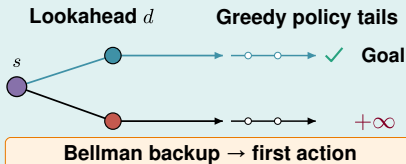
c Controlled HCW transfer



d SOOA graph



e Rollout ADP



f Inspection plan



Rollout ADP with task-aware base-policy evaluation

a Decision state

$$s = (j, m, b, h)$$

j current node
 m covered-target mask
 b selected-view mask
 h remaining budget

Audited + unvisited

$$\mathcal{U}_s(s) : b_a = 0, \chi_{ja} = 1$$

Safe connecting actions retained

Fixed required set $\mathcal{K}$

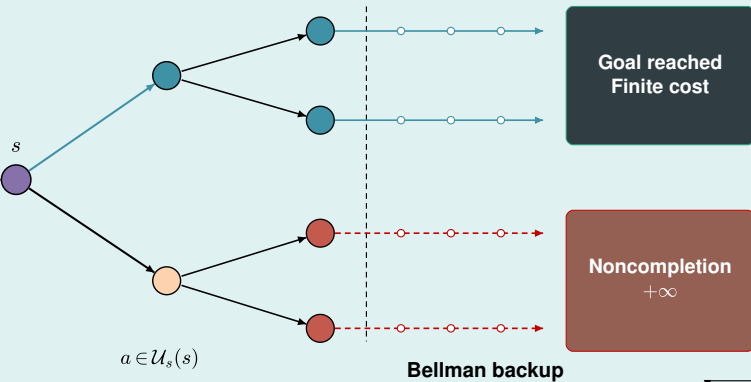
b Rollout ADP

Finite Bellman prefix

Lookahead depth d

Task-aware policy tails

Deterministic greedy μ



All $\widehat{Q}_d(s, a) = +\infty$
 No certified completion

c Policy update

Finite values only

$$a^* = \arg \min_a \widehat{Q}_d(s, a)$$

Append first arc and update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

Complete plan

State update and replan